

1. Server Jalan

```
transactional.id.expiration.ms = 604800000
unclean.leader.election.enable = false
unclean.leader.election.interval.ms = 300000
unstable.api.versions.enable = false
unstable.feature.versions.enable = false
(org.apache.kafka.common.config.AbstractConfig)
[2026-03-10 09:38:20,352] INFO [BrokerServer id=1] Waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2026-03-10 09:38:20,388] INFO [BrokerLifecycleManager id=1] The broker has been unfenced. Transitioning from RECOVERY to RUNNING. (kafka.server.BrokerLifecycleManager)
[2026-03-10 09:38:20,389] INFO [BrokerServer id=1] Finished waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2026-03-10 09:38:20,390] INFO authorizerStart completed for endpoint PLAINTEXT. Endpoint is now READY. (org.apache.kafka.server.network.EndpointReadyFutures)
[2026-03-10 09:38:20,390] INFO [SocketServer listenerType=BROKER, nodeId=1] Enabling request processing. (kafka.network.SocketServer)
[2026-03-10 09:38:20,391] INFO Awaiting socket connections on 0.0.0.0:9092. (kafka.network.DataPlaneAcceptor)
[2026-03-10 09:38:20,392] INFO [BrokerServer id=1] Waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-03-10 09:38:20,392] INFO [BrokerServer id=1] Finished waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-03-10 09:38:20,392] INFO [BrokerServer id=1] Waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-03-10 09:38:20,392] INFO [BrokerServer id=1] Finished waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-03-10 09:38:20,393] INFO [BrokerServer id=1] Transition from STARTING to STARTED (kafka.server.BrokerServer)
[2026-03-10 09:38:20,394] INFO Kafka version: 4.2.0 (org.apache.kafka.common.utils.AppInfoParser)
[2026-03-10 09:38:20,394] INFO Kafka commitId: a18251bae0b825c6 (org.apache.kafka.common.utils.AppInfoParser)
[2026-03-10 09:38:20,394] INFO Kafka startTimeMs: 1773110300393 (org.apache.kafka.common.utils.AppInfoParser)
[2026-03-10 09:38:20,397] INFO [KafkaRaftServer nodeId=1] Kafka Server started (kafka.server.KafkaRaftServer)
```

Buat topik baru

```
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic quickstart-events --bootstrap-server localhost:9092
2026-03-10T02:39:25.769607500Z main ERROR Reconfiguration failed: No configuration found for '2c7b84de' at 'null' in 'null'
Created topic quickstart-events.
```

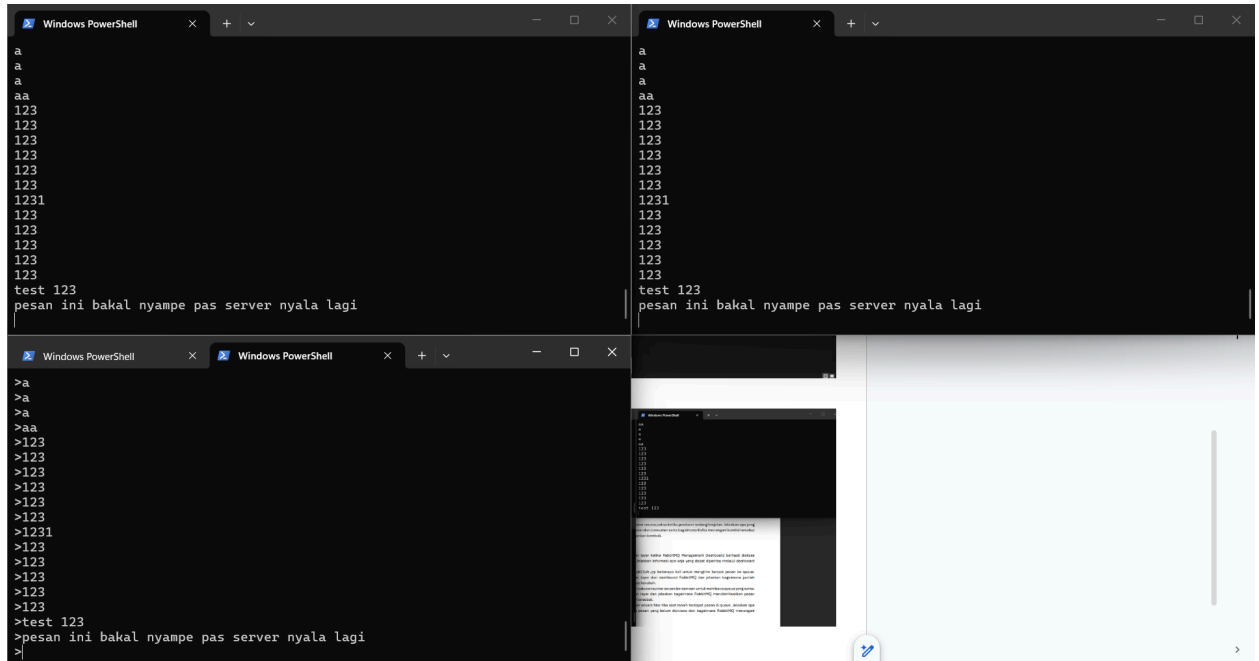
Berdasarkan tangkapan layar, server sudah berjalan ketika ada transisi dari STARTING ke STARTED ketika menjalankan server. Topic baru berhasil terbuat ketika ada pesan “Created topic quickstart-events” muncul di layar.

2. Consumer menerima event

```
PS C:\kafka> .\bin\windows\kafka-console-consumer.bat --topic quickstart-events --from-beginning --bootstrap-server localhost:9092
2026-03-10T02:40:02.035169700Z main ERROR Reconfiguration failed: No configuration found for '2c7b84de' at 'null' in 'null'
test
a
a
a
a
a
a
a
a
a
a
a
a
a
a
a
a
aa
a
a
a
a
a
a
a
a
a
```

Ketika mengirim banyak data, terdapat semacam *delay* yang dapat diobservasi, meskipun *delay*-nya sebentar ketika data diterima oleh consumer. Ketika producer mengirim data dalam jumlah yang besar berterus-terus, maka consumer akan tetap memproses data sesuai dengan kemampuannya sehingga jarak antara data yang dikirim producer dan data yang diproses consumer bisa semakin menjauh. Namun, data yang dikirim tidak akan hilang dan masih akan bisa diproses oleh consumer.

3. Mengirim ke dua consumer



Pada saat server mati, producer tetap mencoba untuk mengirimkan pesan dan consumer akan tetap mencoba menyambung kembali ke server. Namun, pesan yang dikirim saat server mati akan tetap ada di disk sehingga ketika server menyala kembali, producer dan consumer akan secara otomatis mendeteksi server sudah aktif, membaca log file yang ada di disk dan melanjutkan aktivitasnya masing-masing.

5. Informasi Dashboard

- Overview:

health check RabbitMQ, termasuk penggunaan RAM, kapasitas disk, jumlah proses erlang, dan message rate

- Connections:

Daftar aplikasi (producer/consumer) yang terhubung pada RabbitMQ

- Channels:

Memantau saluran dalam sebuah koneksi TCP.

- Exchanges:

Router yang bertugas menerima pesan dari Producer dan mengarahkan ke queue berdasarkan routing

- Queues and Streams:

Tempat untuk memantau queue, dapat dilihat statusnya, yaitu ready dan unacked serta total pesan dari tiap queue

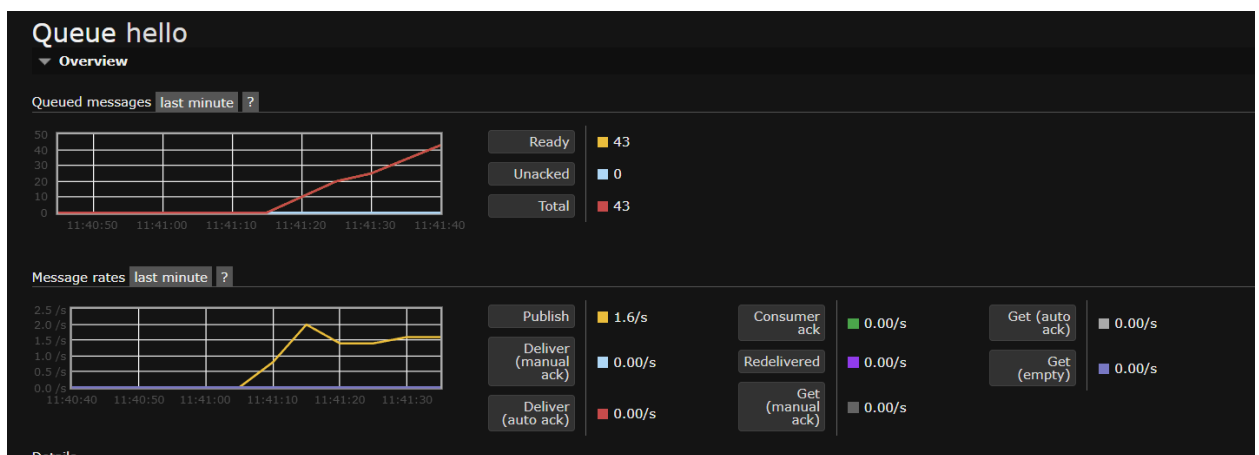
- **Admin:**

Tempat mengelola users dan hak akses.

Referensi:

<https://www.rabbitmq.com/docs>

6. Menjalankan [publish.py](#)



Grafik Ready dan Total akan naik seiring [publish.py](#) dijalankan. Ketika producer mengirim pesan tanpa ada yang membaca, RabbitMQ berperan sebagai *buffer* dan disimpan sampai ada consumer yang terhubung.

7. Distribusi pesan

mengganti status menjadi ready secara otomatis. Jadi, Ketika consumer sudah menyala Kembali, pesan yang ready akan dikirimkan ke consumer dan akan muncul pada layar.

9. Memastikan database berjalan bersama

```
Administrator: Windows PowerShell
mysql> CREATE USER 'replica'@'%' IDENTIFIED WITH mysql_native_password BY 'password';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT REPLICATION SLAVE ON *.* TO 'replica'@'%';
Query OK, 0 rows affected (0.00 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW MASTER STATUS;
+-----+-----+-----+-----+-----+
| File           | Position | Binlog_Do_DB | Binlog_Ignore_DB | Executed_Gtid_Set |
+-----+-----+-----+-----+-----+
| mysql-bin.000001 | 833      |               |                   |                   |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

```
Administrator: Windows PowerShell
mysql> SHOW SLAVE STATUS\G
***** 1. row *****
Slave_IO_State: Waiting for source to send event
Master_Host: 127.0.0.1
Master_User: replica
Master_Port: 3306
Connect_Retry: 60
Master_Log_File: mysql-bin.000001
Read_Master_Log_Pos: 833
Relay_Log_File: relay-bin.000002
Relay_Log_Pos: 326
Relay_Master_Log_File: mysql-bin.000001
Slave_IO_Running: Yes
Slave_SQL_Running: Yes
Replicate_Do_DB:
Replicate_Ignore_DB:
Replicate_Do_Table:
Replicate_Ignore_Table:
Replicate_Wild_Do_Table:
Replicate_Wild_Ignore_Table:
Last_Errno: 0
Last_Error:
Skip_Counter: 0
Exec_Master_Log_Pos: 833
Relay_Log_Space: 530
Until_Condition: None
Until_Log_File:
Until_Log_Pos: 0
Master_SSL_Allowed: No
```

```
PS C:\Users\Fathurrahman> netstat -ano | Select-String "3306|3307"

TCP    0.0.0.0:33060          0.0.0.0:0           LISTENING        21244
TCP    0.0.0.0:33060          0.0.0.0:0           LISTENING        38696
TCP    127.0.0.1:3306        0.0.0.0:0           LISTENING        38696
TCP    127.0.0.1:3306        127.0.0.1:19466     ESTABLISHED      38696
TCP    127.0.0.1:3306        127.0.0.1:21459     ESTABLISHED      38696
TCP    127.0.0.1:3307        0.0.0.0:0           LISTENING        21244
TCP    127.0.0.1:3307        127.0.0.1:6311      ESTABLISHED      21244
TCP    127.0.0.1:6311        127.0.0.1:3307      ESTABLISHED      15388
TCP    127.0.0.1:19466       127.0.0.1:3306      ESTABLISHED      21244
TCP    127.0.0.1:21459       127.0.0.1:3306      ESTABLISHED      43512
TCP    [::]:33060            [::]:0              LISTENING        38696
TCP    [::]:33060            [::]:0              LISTENING        21244
```

Master dan Slave berhasil dijalankan. Melakukan login ke database dan melakukan netstat pada port 3306 dan 3307 juga memastikan bahwa kedua database tersebut sudah berjalan secara bersamaan.

10. Insert di Master

```
PS C:\Program Files\MySQL\MySQL Server 8.0\bin> ./mysql -u root -h 127.0.0.1 -P 3306
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use testdb;
Database changed
mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO users VALUES(2,'Test');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO users VALUES(3,'Test');
Query OK, 1 row affected (0.00 sec)

mysql>
```

```
+-----+
2 rows in set (0.00 sec)

mysql> use testdb;
Database changed
mysql> delete from users where id ==2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '==2' at line 1
mysql> delete from users where id =2;
Query OK, 1 row affected (0.05 sec)

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
+-----+
1 row in set (0.00 sec)

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
+-----+
3 rows in set (0.00 sec)

mysql>
```

Insert di master juga muncul di slave

11. Mematikan Master

```
Administrator: Windows PowerShell
Your MySQL connection id is 10
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use testdb;
Database changed
mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO users VALUES(2,'Test');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO users VALUES(3,'Test');
Query OK, 1 row affected (0.00 sec)

mysql> exit
bye
PS C:\Program Files\MySQL\MySQL Server 8.0\bin>
```

```
Administrator: Windows PowerShell

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
+-----+
1 row in set (0.00 sec)

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
+-----+
3 rows in set (0.00 sec)

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
+-----+
3 rows in set (0.00 sec)

mysql>
```

Ketika master dimatikan, Slave masih bisa diakses dan masih bisa melakukan query di testdb. Hal ini bisa terjadi karena adanya replikasi ke database Slave.

12. Insert di Slave

```
Administrator: Windows PowerShell

mysql> exit
bye
PS C:\Program Files\MySQL\MySQL Server 8.0\bin> ./mysql -u root -h 127.0.0.1 -P 3306
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select * from users;
ERROR 1046 (3D000): No database selected
mysql> use testdb;
Database changed
mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
+-----+
3 rows in set (0.00 sec)

mysql>
```

```
Administrator: Windows PowerShell

mysql> select * from users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO users VALUES(2, 'User_Dari_Slave');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM users;
+-----+
| id | name |
+-----+
| 1 | Test |
| 2 | Test |
| 3 | Test |
| 2 | User_Dari_Slave |
+-----+
4 rows in set (0.00 sec)

mysql>
```


Operasi query insert di Slave hanya berlaku pada Slave dan tidak muncul pada database Master karena sifat Master Slave hanya berlaku satu arah, yaitu dari Master ke Slave.